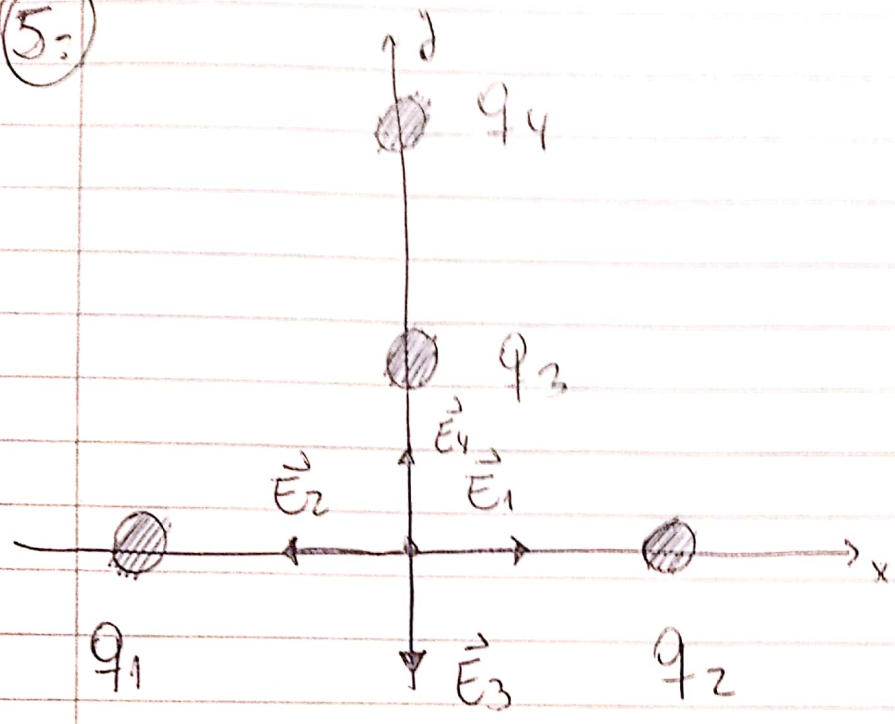
5:



$$\vec{E}_1 = K \frac{q_2}{d^3} d\hat{\imath}$$

$$\vec{E}_2 = K \frac{q_1}{d^3} d(-\hat{\imath})$$

$$\vec{E}_3 = K \frac{q_3}{d^3} d(-\hat{\jmath})$$

$$\vec{E}_4 = K \frac{q_4}{(2d)^3} (2d) \hat{\jmath}$$

Escaneado con CamScanner

$$\vec{E} = \sum_{i=1}^{4} \vec{E}_i$$

$$= \frac{K}{d^2}\left[q_2 \hat{\imath} + q_1(-\hat{\imath}) + q_3(-\hat{\jmath}) + \frac{q_4}{4}\hat{\jmath} \right]$$

$$= \frac{9 \cdot 10^9}{(0{,}005)^2}\left[\overset{\to 0}{\cancel{(q_2 - q_1)}}\hat{\imath} + 15 \cdot 10^{-6}(-\hat{\jmath}) + 15 \cdot 10^{-6}\hat{\jmath} \right]$$

$$= \vec{0}.$$

Escaneado con CamScanner